



**SILVER OAK  
UNIVERSITY**  
EDUCATION TO INNOVATION

**School of Technology, Design and computer Application**  
**Silver oak College of computer Application**  
**Department of computer science**  
**Bachelor of Science Computer Science and Information Technology**

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|---------------------|-------------------------|-----------------------|------------|
| <b>Semester:</b>    | 5                       | <b>Academic Year:</b> | 2025-26    |
| <b>Course Name:</b> | Mobile Game Development | <b>Course Code:</b>   | 3040243338 |

**Question Bank**

| Sr. No.                                                 | Question Text                                                                          | Marks | CO   |
|---------------------------------------------------------|----------------------------------------------------------------------------------------|-------|------|
| <b>UNIT 1 : Introduction to Mobile Game Development</b> |                                                                                        |       |      |
| 1.                                                      | Explain what is meant by mobile game development.                                      | 2     | CO-1 |
| 2.                                                      | Describe any two popular trends in mobile gaming.                                      | 2     | CO-1 |
| 3.                                                      | What do you understand by game mechanics?                                              | 2     | CO-1 |
| 4.                                                      | List two differences between Unity and Godot game engines.                             | 2     | CO-1 |
| 5.                                                      | Briefly define the pre-production phase in game development.                           | 2     | CO-1 |
| 6.                                                      | Describe the three key components of game design: mechanics, dynamics, and aesthetics. | 3     | CO-1 |
| 7.                                                      | Explain the role of a game engine in mobile game development.                          | 3     | CO-1 |
| 8.                                                      | Outline the production phase in the mobile game development pipeline.                  | 3     | CO-1 |
| 9.                                                      | Compare mobile game genres with examples.                                              | 3     | CO-1 |
| 10.                                                     | Summarize the importance of studying game development history.                         | 3     | CO-1 |
| 11.                                                     | Explain how mobile game trends have evolved over the past decade.                      | 4     | CO-1 |
| 12.                                                     | Describe the major differences between Unity, Unreal Engine, and Godot.                | 4     | CO-1 |



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|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|---|------|
| 13.                                           | Discuss the three stages of the game development pipeline with appropriate examples.                                          | 4 | CO-1 |
| 14.                                           | How do mechanics, dynamics, and aesthetics together contribute to good game design?                                           | 4 | CO-1 |
| 15.                                           | Explain with a case study how a specific mobile game represents its genre effectively.                                        | 4 | CO-1 |
| 16.                                           | Explain the complete mobile game development process from concept to post-production.                                         | 5 | CO-1 |
| 17.                                           | Describe in detail how Unity and Godot differ in their usability and target audience.                                         | 5 | CO-1 |
| 18.                                           | Analyze the evolution of mobile games and how market demand has influenced game design.                                       | 5 | CO-1 |
| 19.                                           | Explain the significance of game design principles (mechanics, dynamics, aesthetics) in creating an engaging game experience. | 5 | CO-1 |
| 20.                                           | Discuss different mobile game genres with relevant examples and identify key elements that define each genre.                 | 5 | CO-1 |
| <b>UNIT 2 : Game Development Fundamentals</b> |                                                                                                                               |   |      |
| 21.                                           | Explain gravity in game physics in simple terms.                                                                              | 2 | CO-2 |
| 22.                                           | How is KinematicBody2D different from RigidBody2D?                                                                            | 2 | CO-2 |
| 23.                                           | Describe what a sprite is with one example.                                                                                   | 2 | CO-2 |
| 24.                                           | State how textures are useful in game development.                                                                            | 2 | CO-2 |
| 25.                                           | Why are sound effects important in games?                                                                                     | 2 | CO-2 |
| 26.                                           | Define audio optimization in mobile games.                                                                                    | 2 | CO-2 |
| 27.                                           | How do collisions affect RigidBody2D in a game?                                                                               | 3 | CO-2 |
| 28.                                           | Explain the steps to animate a sprite in a 2D game.                                                                           | 3 | CO-2 |
| 29.                                           | Differentiate between sprites and textures with examples.                                                                     | 3 | CO-2 |
| 30.                                           | Describe how the move_and_slide() function is used in KinematicBody2D.                                                        | 3 | CO-2 |



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|----------------------------------------------------|--------------------------------------------------------------------------------------------------------|---|------|
| 31.                                                | Explain the role of Panels and Buttons in mobile game UI.                                              | 3 | CO-2 |
| 32.                                                | Give an example of how a HUD is used in a mobile game.                                                 | 3 | CO-2 |
| 33.                                                | Explain the importance of menus in mobile game UI.                                                     | 3 | CO-2 |
| 34.                                                | Explain the importance of touch controls in mobile games.                                              | 3 | CO-2 |
| 35.                                                | Describe one use each of an accelerometer and gyroscope in gaming.                                     | 3 | CO-2 |
| 36.                                                | Describe how textures are applied to 3D models.                                                        | 4 | CO-2 |
| 37.                                                | Explain why animations are important in mobile games.                                                  | 4 | CO-2 |
| 38.                                                | Explain with an example how gyroscope enhances the mobile gaming experience.                           | 4 | CO-2 |
| 39.                                                | How do Labels and Panels improve user experience in mobile game UI?                                    | 4 | CO-2 |
| 40.                                                | Explain the role of gravity and collisions in mobile game physics with examples.                       | 5 | CO-2 |
| 41.                                                | Explain with examples how sprites, textures, and animations are used in mobile games.                  | 5 | CO-2 |
| 42.                                                | Describe the steps of integrating 3D models into a mobile game.                                        | 5 | CO-2 |
| 43.                                                | Explain the design principles of UI/UX in mobile games using Control nodes.                            | 5 | CO-2 |
| 44.                                                | Explain background music, sound effects, and audio optimization in detail.                             | 5 | CO-2 |
| 45.                                                | Why is audio optimization a challenge in mobile games? Explain with solutions.                         | 5 | CO-2 |
| 46.                                                | Describe in detail touch, accelerometer, and gyroscope controls with examples.                         | 5 | CO-2 |
| 47.                                                | Explain the design of a sample mobile game UI that includes HUD, menus, buttons, and background music. | 5 | CO-2 |
| <b>UNIT 3 : Advanced Game Development Concepts</b> |                                                                                                        |   |      |
| 48.                                                | What is the role of Path2D in pathfinding?                                                             | 2 | CO-3 |



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|-----|----------------------------------------------------------------------------------------------------------------------|---|------|
| 49. | Differentiate between Navigation2D and Path2D in Godot.                                                              | 2 | CO-3 |
| 50. | What is Peer-to-Peer networking in multiplayer games?                                                                | 2 | CO-3 |
| 51. | State one advantage of using ENet in multiplayer games.                                                              | 2 | CO-3 |
| 52. | Explain how NPC behavior can be managed using state machines.                                                        | 3 | CO-3 |
| 53. | Describe the basic working of Pathfinding in games.                                                                  | 3 | CO-3 |
| 54. | Compare Peer-to-Peer and Server-Based multiplayer systems.                                                           | 3 | CO-3 |
| 55. | Why is Navigation2D important for 2D game AI movement?                                                               | 3 | CO-3 |
| 56. | Illustrate with an example how Godot's High-Level Multiplayer API simplifies networking.                             | 3 | CO-3 |
| 57. | Explain with a diagram how a state machine controls NPC actions (e.g., Idle, Attack, Patrol).                        | 4 | CO-3 |
| 58. | How does Path2D help in defining and following a path in 2D games?                                                   | 4 | CO-3 |
| 59. | Differentiate between ENet and Godot High-Level Multiplayer API.                                                     | 4 | CO-3 |
| 60. | Describe the challenges of using Peer-to-Peer networking in games.                                                   | 4 | CO-3 |
| 61. | Explain the importance of combining State Machines and Pathfinding for intelligent NPCs.                             | 4 | CO-3 |
| 62. | Discuss in detail how a finite state machine can improve NPC decision-making in games.                               | 5 | CO-3 |
| 63. | Explain the complete process of implementing pathfinding using Path2D and Navigation2D in Godot.                     | 5 | CO-3 |
| 64. | Compare Peer-to-Peer and Server-Based networking with their advantages and disadvantages in multiplayer game design. | 5 | CO-3 |
| 65. | How does ENet handle latency and reliability in multiplayer games?                                                   | 5 | CO-3 |
| 66. | Write a detailed explanation of how AI and Multiplayer concepts together enhance the gaming experience.              | 5 | CO-3 |